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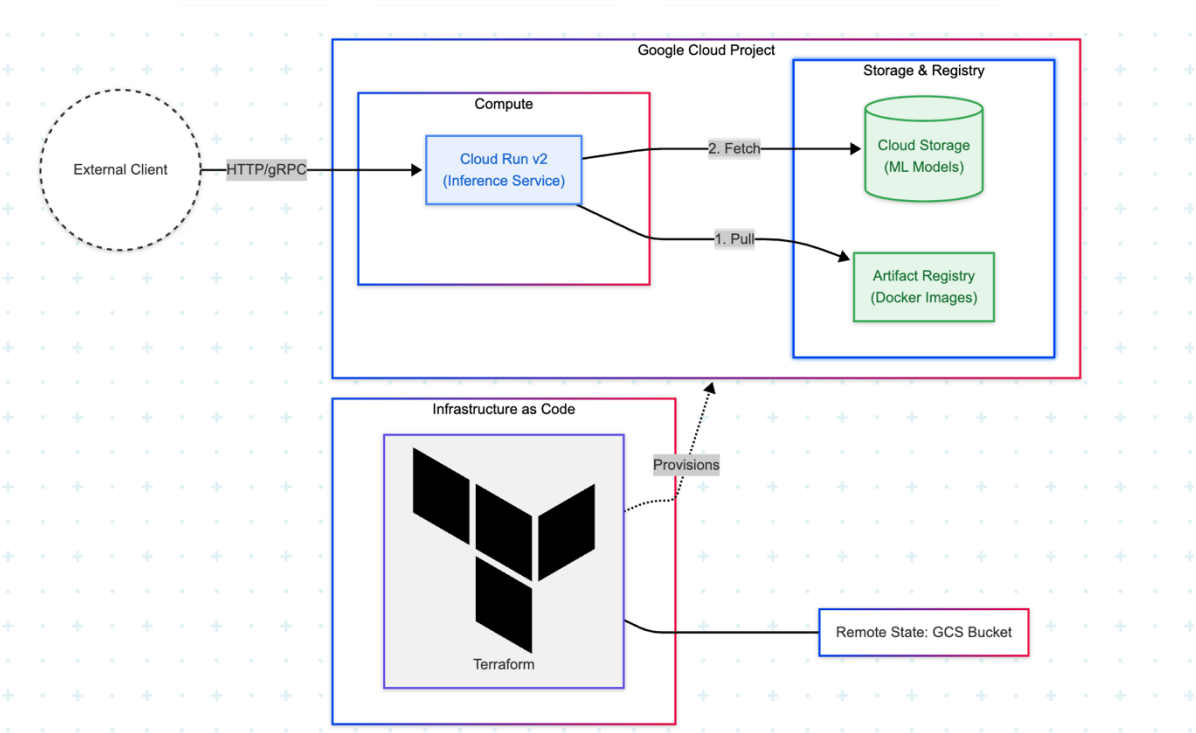
Project: Agentic AI Powered Cloud Infrastructure Automation for ML Inference

Overview

I designed and deployed a fully automated cloud infrastructure for machine learning inference using Terraform, enhanced with Agentic AI to accelerate development, reduce errors, and drastically shorten deployment time.

What traditionally required a team effort over several weeks was completed within hours by combining Infrastructure as Code with an AI agent capable of interacting directly with my development environment.

Architecture



- **Cloud Run** → serves ML inference
- **GCS Bucket** → stores trained ML models
- **Artifact Registry** → stores container images
- **Service Account + IAM** → secure access between services
- **Terraform Modules** → reusable infra definitions
- Google Cloud Run (GPU enabled, NVIDIA L4)
- Docker
- Agentic AI (workspace-integrated

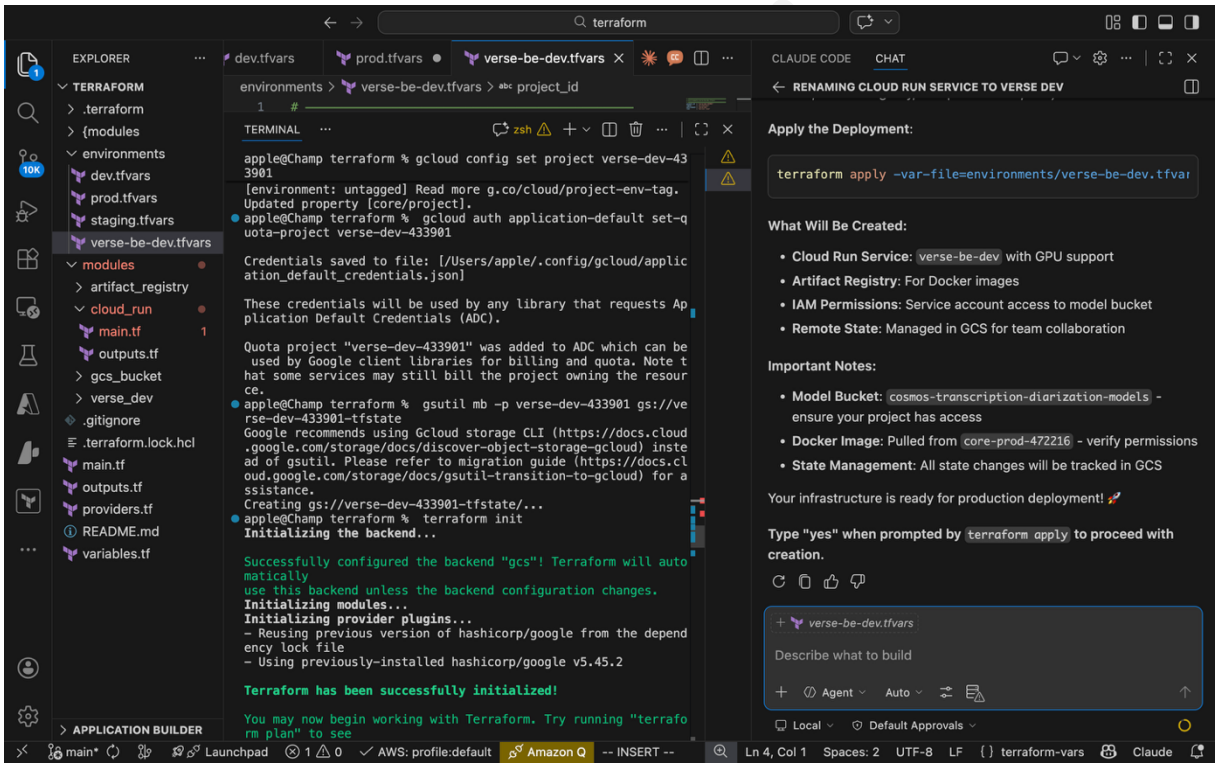
Flow:

1. Model is stored in GCS bucket
2. Cloud Run service reads model path via config
3. Container loads model at runtime
4. Inference requests hit Cloud Run endpoint
5. Results returned in real time

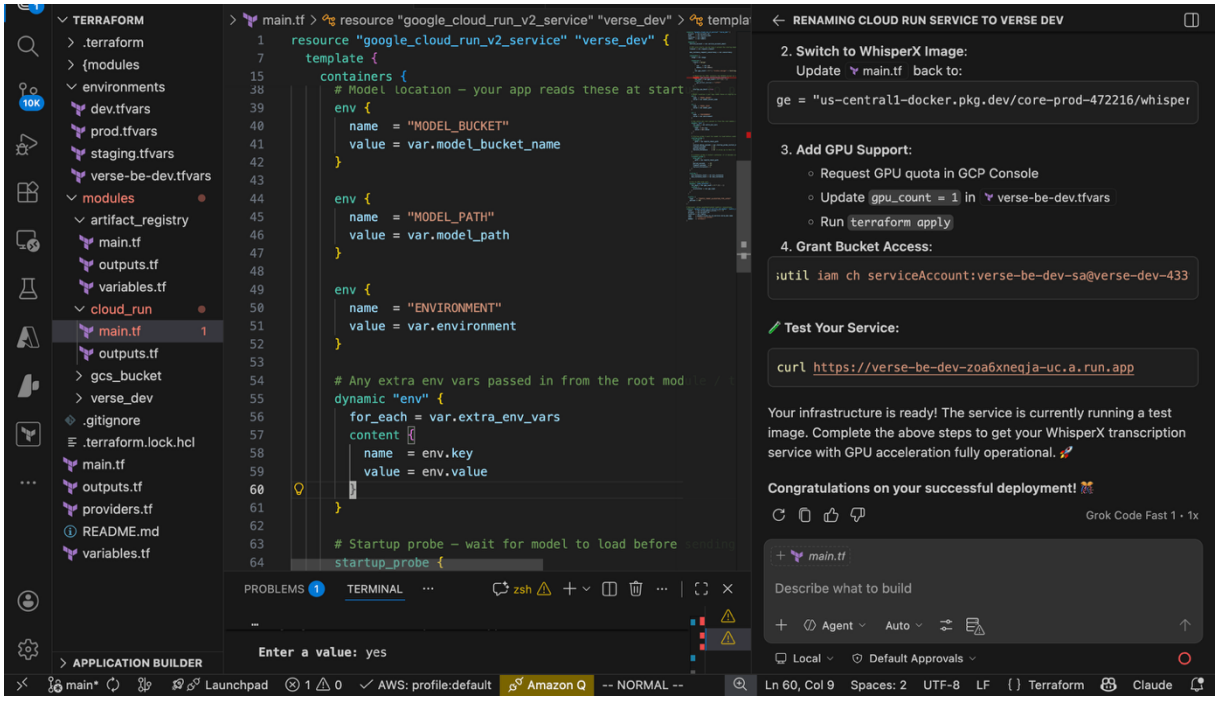
Where Agentic AI Comes In

Instead of just "using AI for suggestions", I used it as an **active operator (Junior Engineer)**:

- Read and analyzed Terraform files
- Modified infrastructure code in real time
- Removed deprecated variables (MODEL_BUCKET, MODEL_PATH, ENVIRONMENT)



- Implemented GPU configuration (NVIDIA L4)
- Executed Terraform commands:
  - terraform plan
  - terraform apply



- Interpreted errors (like GPU zonal redundancy issue)
- Applied fixes (disabled zonal redundancy)
- Validated deployment success via endpoint testing

This is **Agentic AI behaviour**.

⚡ Key Innovation

I leveraged an AI coding agent as an execution layer within my workflow, enabling it to

- Operate directly on my workspace rather than just providing suggestions
- Execute a continuous loop of **edit → run → debug → fix → validate**
- Reduce context switching between development, deployment, and debugging
- Eliminate manual trial-and-error cycles during infrastructure setup

In addition, I designed a seamless inference workflow where:

- Newly uploaded models in Cloud Storage are automatically referenced at runtime
- Cloud Run dynamically retrieves the latest model path
- Updated models are served without requiring full service redeployment

Impact

- Reduced deployment time from **2–4 weeks → a few hours**
- Faster debugging and iteration cycles
- Enabled rapid infra changes without breaking environments
- Scaled individual productivity to **team-level output**

By Ikenna Anaziezo